



Sensorimotor synchronization with visual, auditory, and tactile modalities

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Abstract

While it is well known that humans are highly responsive to rhythm, the factors that influence our ability to synchronize remain unclear. In the current study, we examined how stimulus modality and rhythmic deviation, along with the synchronizer's level of musicality, impacted sensorimotor synchronization (SMS). Utilizing a finger-tapping task and three sensory modalities (visual, auditory, and tactile), we manipulated rhythmic deviation by varying the temporal position, intensity, and availability of cues across four deviation levels. Additionally, to determine our participants' musical familiarity and aptitude, we administered the Goldsmiths Musical Sophistication Index (Gold-MSI) questionnaire. We found that SMS to external rhythmic stimuli was significantly more precise for auditory and tactile than for visual sequences. Further, we found SMS consistency significantly decreased in all modalities with increased rhythmic deviation, suggesting rhythmic deviation directly relates to SMS difficulty. Moreover, a significant correlation was found between Gold-MSI scores and SMS consistency in the most rhythmically deviant level, such that the higher one's musical general sophistication score, the greater one's SMS ability. This held for all three modalities. Combined, these findings suggest that rhythmic synchronization performance is affected not only by the modality and rhythmic deviation of the stimuli but also by the musical general sophistication of the synchronizer.

Introduction

Sensorimotor synchronization (SMS), the coordination of physical movement in time with a rhythmic sequence, can be witnessed overtly with dancing and tapping to the tempo of music (D'Andrea-Penna et al., 2020). SMS requires the active perception of an external (or imagined) stimulus and the preparation for movement to it (Repp, 2005). Neural connections between sensory and motor processes are acutely developed, and serve sensorimotor communicative abilities via the detection and articulation of body movement, gestures, and spoken language (Pezzulo et al., 2019). SMS is essential for referencing and storage of knowledge, and basic brain function (Hawkins et al., 2017), and is of flourishing interest in the fields of neuroscience and human–robot communication (Navarro-Alarcon et al., 2020). Humans' high responsiveness to rhythm has presented the opportunity to study SMS with discrete stimuli, without the complexities

of semantics and language. However, factors that influence our ability to synchronize are not well understood (Schiavio et al., 2015).

The modality of the stimulus is reported to be an important factor, due to the superior temporal saliency of audition and the tendency to associate rhythm with music (e.g., Bartlett & Bartlett, 1959; Grahn, 2012; Repp & Penel, 2002; Su, 2016). Tapping studies investigating differences in SMS abilities between auditory and tactile modalities have found similar performance in terms of accuracy (i.e., average relative phase) and consistency (i.e., the standard deviation of relative phase) of tap timing when steady sequences were used (Brochard et al., 2008). However, when subdivisions of the beat were added to the cue sequence, auditory performance was more consistent (with a lower standard deviation of relative phase), indicating better SMS for auditory stimuli (Ammirante et al., 2016). While the consistency of tapping across a time series offers a broad indication of the ability to synchronize (Nozaradan et al., 2016), accuracy can be taken to represent the sensory ability to detect a steady beat (see Repp & Su, 2013 for a review). Large individual differences in SMS accuracy have been reported (Aschersleben, 2003). When individuals synchronize to a steady beat with a

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fixed inter-stimulus-interval (e.g., 600 ms), accuracy values can range from negative (e.g., -300 ms; tapping before) to positive (e.g., $+300$ ms; tapping after), with zero indicating precise tapping on the beat. Negative accuracy values are commonly found in non-musicians tapping to a steady auditory metronome and suggest a tendency to anticipate the following tone (Repp, 2005). Even after receiving training that aimed to minimize tapping errors, participants reported difficulty in distinguishing the objective experience of when the beats occurred from their anticipatory tendency (Ascherleben, 2003).

Generally, auditory and tactile modalities elicit better synchronization than the visual modality (see Repp & Su, 2013 for a review). SMS with discrete visual flashes was significantly less consistent than with auditory tones (Repp & Penel, 2004). This result was replicated, and findings suggest perception–action coupling might differ between vision and audition (Patel et al., 2005). However, the opposite has also been reported, with better SMS performance for visual stimuli than for auditory tones when a continuous visual stimulus that incorporates both temporal and spatial components was used (Armstrong & Issartel, 2014). Other studies that employed a bouncing ball stimulus (Gan et al., 2015; Iversen et al., 2015) or a video of finger-tapping (Hove & Keller, 2010) reported similar SMS consistency between visual and auditory modalities, suggesting that the addition of spatial information with a discrete (collision) point facilitates better synchronization in the visual modality. When a multi-sensory stimulus is used (e.g., audio-visual cues), more consistent SMS performance can be achieved than when using a uni-sensory stimulus (Armstrong & Issartel, 2014). The same effect has been reported for auditory-tactile and tactile-visual cues (Elliott et al., 2010). Synchronizers tend to rely more on auditory signals over visual and tactile ones, and more on tactile when auditory signals are absent (Elliott et al., 2010).

The time-keeping strategies used to achieve SMS consistency can be revealed by the autocorrelation structure of the tapping time course (Iversen et al., 2015). Autocorrelation can describe how the timing of a tap is related to the timing of its preceding taps. A tapping time course reflects a series of dynamic statistical predictions of external temporal events, potentially providing insight into underlying timing mechanisms (Vorberg & Wing, 1996). The correlation between one tap and the previous tap is known as lag-1 or autocorrelation-1 (AC-1). For SMS tasks where the stimulus has a simple one-beat pattern, it is common to observe only significant positive or negative AC-1, as opposed to significant AC values at higher-order lags indicative of the metrical time-keeping strategy for beat patterns that contain two or more beats (Ammirante et al., 2016). Significantly negative AC-1 reflects predictive error correction, e.g., realizing the last tap was slightly ahead, and purposely adjusting by

delaying the next tap in an anticorrelated manner. Error correction is commonly associated with SMS to auditory tones (Guérin et al., 2021; Iversen et al., 2015) and tactile vibrations (Ammirante et al., 2016). Therefore, a significantly negative AC-1 is indicative of a predictive timing strategy, most commonly used in finger-tapping tasks, where motor coordination to an explicit rhythmic event is required (Ivry et al., 2002). On the other hand, positive AC-1 is associated with implicit drift in and out of synchrony as the time series unfolds, a consequence of emergent kinetic features of tapping movement (Zelaznik et al., 2002). In other words, emergent timing is characterized by less reliance on the explicit prediction of the next stimulus, and instead on the continuity of action (Ivry et al., 2002). A significantly positive AC-1 is a reported feature of SMS to visual flashes and suggests generally weaker entrainment to the external stimulus (Hove et al., 2010). Iversen et al. (2015) found that tapping to a bouncing ball evoked a similar negative AC-1 to that of tapping to tones, suggesting that the presence of visual motion could compensate for the inferior temporal saliency of visual flashes.

Another factor that influences SMS performance is cue availability within the pattern (Kung et al., 2013). Adding temporally congruent subdivisions between beats has been shown to increase the accuracy (Repp et al., 2005) and consistency (Ammirante et al., 2016) of SMS performance. On the other hand, SMS performance can be worsened by “increasing negative evidence” of the beat, i.e., increasing rhythmic deviation from the pattern participants are tasked to follow (McAuley & Semple, 1999). Bouwer et al. (2018) found that either removing beats or adding temporally incongruent subdivisions worsened synchronization performance. Adding temporally incongruent subdivisions makes synchronization challenging by introducing emphasis that deviates from the beat (Bouwer et al., 2018; Maes, 2016). The presence of an additional $\frac{1}{2}$ note before the main cue (in musical jargon, an ‘anacrusis’ or ‘pick-up’ due to it being followed by an expected main cue), or after the main cue (an ‘acephalic’ due to the lack of the subsequently expected main cue) both increase deviation from the underlying beat (London, 2012). Interestingly, while an ‘anacrusis’ or ‘acephalic’ cue is typically distracting and could impair SMS performance, their presence may assist SMS performance in those with a sufficient level of musicality by stimulating alternative synchronization strategies (Sioros et al., 2018).

Last but not least, the synchronizer’s musicality can affect SMS performance. Here, we defined musicality as a broad and complex measure of musical familiarity and aptitude. Varying degrees of musicality are proposed as a source of variability in SMS performance between participants (Repp, 2005). Music training, often in terms of years taught on a musical instrument, is a commonly used measure of musical aptitude (Białuńska & Dalla Bella, 2017; Chen et al.,

2002). However, it is our view that other critical features of musicality have been overlooked. The Goldsmiths Musical Sophistication Index (Gold-MSI) questionnaire (Müllensiefen et al., 2014) offers a more comprehensive approach to assessing musicality in individuals who would not necessarily consider themselves highly musical. By including additional assessments of active engagement, perceptual abilities, emotions, and singing abilities, Gold-MSI musical general sophistication scores might serve as better covariates than one's level of music training when modeling SMS behavioral data in studies involving rhythm production (Matthews et al., 2016; Stupacher, 2019).

In the current study, we examined how manipulation of rhythmic deviation and modality, along with the synchronizer's level of musicality, impacted SMS. Utilizing a finger-tapping task and three sensory modalities (auditory, tactile, and visual), we manipulated rhythmic deviation by varying the location and availability of temporal cues across four deviation levels ('full-time', 'half-time', 'half + anacrusis', and 'half + acephalic'). Additionally, we administered the Gold-MSI questionnaire to determine our participants' musical familiarity and aptitude. We predicted: i) SMS performance would be better for auditory and tactile modalities than for visual; ii) SMS performance in all three modalities would decrease as rhythmic deviation increased; and iii) Those with higher scores on the Gold-MSI would have better SMS performance.

Experimental methods

Participants

30 right-handed participants (15 women, age $M = 24.4$, $SD = 6.2$) were included for this study. Our study was approved by the Institution Review Board of the University of Nevada, Reno, and conducted in accordance with the Code of Ethics of the World Medical Association (Declaration of Helsinki). All participants reported having normal hearing and normal or corrected-to-normal vision. Participants were compensated for their participation in one 90-min session.

Stimuli

We used separate visual, auditory, or tactile stimuli to create a metronomic beat with a tempo of 100 beats per minute (bpm), and an inter-stimulus-interval (ISI) of 600 ms. This is within the range of preferred spontaneous motor tempi and was previously found to be easy to synchronize with (Guérin et al., 2021). Visual and auditory stimuli were generated using MATLAB and PsychToolbox (Brainard, 1997; Pelli, 1997). Visual stimuli were 33 ms black circles (3° or 1.5°

diameter) presented centrally on a white background with a refresh rate of 60 Hz. Auditory stimuli were 30 ms pure tones (with a 5 ms rise and fall) of 500 Hz (60 dB or 50 dB) delivered through headphones (Sennheiser HD 250). The visual and auditory stimuli were delivered through a Display++ system with a refresh rate of 120 Hz and an AudioFile stimulus processor (Cambridge Research Systems, Rochester, UK). Tactile stimuli were 30 ms vibrations at 50 Hz (1.2 mm or 0.6 mm peak-to-peak displacement) generated with PiezeTac tactor (Engineering Acoustic, Florida, USA) and delivered to participants' non-dominant left index finger.

Behavioral finger-tapping task procedure

To measure the participant's SMS abilities across three sensory modalities (auditory, tactile, and visual), a finger-tapping task was employed. Participants sat at a desk 60 cm in front of the LCD monitor and a keyboard (HHKB lite 2) with measured latency of 20 ms (factored into raw data recording). Participants were instructed to tap the space bar on the keyboard with their dominant index finger in synchrony with the steady meter (tempo) defined by flashes presented on the monitor (visual trials), tones presented via headphones (auditory trials), or vibrations delivered on their left index finger (tactile trials). Participants wore headphones throughout all trials to prevent external sounds from interfering with the task.

Each trial started with an introduction phase, during which participants were instructed to start tapping and get into synchrony. Each tap of the introduction phase had a coincident cue. The length of the introduction was randomly chosen between 7 and 12 beats to prevent the use of a counting strategy. The introduction phase also served the purpose of minimizing data rejection due to initial errors. In a previous synchronization study, the first four taps were discarded from every trial due to initial performance error (Iversen et al., 2015). The introduction phase was followed by 50 beats, during which participants were tasked to tap in synchrony with the stimuli as accurately as possible until the trial ended.

The tempo that participants were required to tap remained constant at 100 bpm. We manipulated rhythmic deviation by varying the temporal position and availability of cues across four deviation levels (Fig. 1a). In the 'full-time' deviation level (with no rhythmic deviation) every tap had a coincident cue. In the 'half-time' deviation level, only every other tap had a coincident cue, effectively halving the cue availability to that of a 50-bpm beat, while keeping tapping at 100 bpm. Participants were told that the target tempo remained the same, but they were being given only half of the cues relative to the previously performed 'full-time' condition. This removal of timing framework information increased

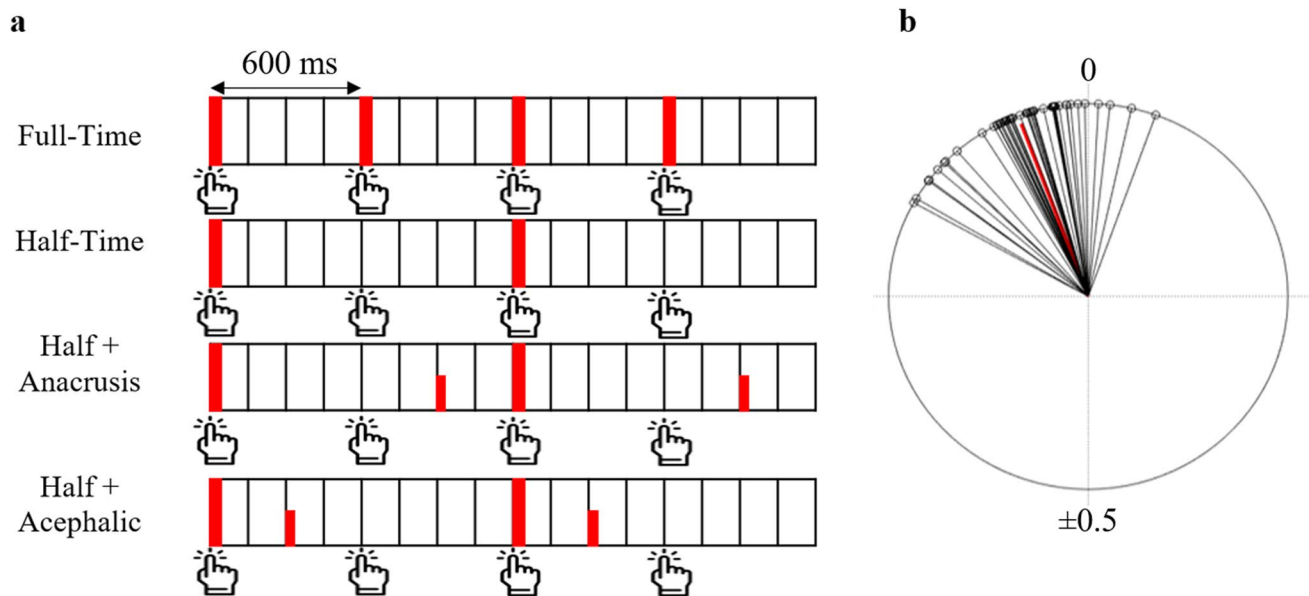


Fig. 1 The four finger-tapping synchronization deviation levels. **a** Each row illustrates four example taps in one of four deviation levels. The red markers represent cues. Tall red markers represent full-intensity main cues and short ones represent half-intensity additional cues. **b** An example of a 50-tap trial with accuracy vectors for all taps

(black lines) and mean accuracy vector (red line). Accuracy value zero represents a perfectly accurate tap on the beat, whereas ± 0.5 represents the most inaccurate tap before (negative) or after (positive) the target beat (color figure online)

the possibility of uncertainty (and therefore prediction error) around tap synchrony. In the ‘half + anacrusis’ and ‘half + acephalic’ deviation levels, while there was still the missing cue (as in the ‘half-time’ deviation level), an additional cue was added to further increase rhythmic deviation and task difficulty. The additional cue was presented at half intensity (i.e., 1.5° diameter flash for visual, 50 dB tone for auditory, and 0.6 mm displacement for tactile) of the main cue (i.e., 3° diameter flash for visual, 60 dB tone for auditory, and 1.2 mm displacement for tactile). The extra cue was presented as a $\frac{1}{2}$ note (300 ms) before (‘half + anacrusis’ deviation level) or after the main cue (‘half + acephalic’ deviation level). The latter presentation of the cue made the deviation level the most difficult out of the four due to it further emphasizing the absence of the expected cue (Kiparsky, 2018). Note that in all three ‘half’ deviation levels, our manipulations of rhythmic deviation also caused the pattern to reset every two beats, rather than every beat (as was the case in the ‘full-time’ deviation level).

Visual, auditory, and tactile trials were presented in separate sessions, with the order of sessions counterbalanced across participants. In each session, there were 6 trials for each of the four deviation levels, resulting in a total of 24 trials. The deviation levels within the session were presented in a fixed order, from ‘full-time’ to most rhythmically deviant ‘half + acephalic’ (Fig. 1a). Information describing which deviation level the next trial would be, and a reminder of task instructions was displayed before each trial. Participants

started each trial at their own pace by pressing the spacebar on the keyboard. When participants completed all 6 trials of one deviation level, they then moved on to the next. Each trial lasted approximately 40 s. The total running time for each session was approximately 20 min, and participants were given five minutes break between sessions. The experiment lasted approximately 75 min.

Musicality self-report questionnaire

Each participant completed the Goldsmiths Musical Sophistication Index v.1.0 (Gold-MSI) 38-item questionnaire (Müllensiefen et al., 2014). Each item was presented as a statement, e.g., “I spend a lot of my free time doing music-related activities.” and responses were given using an agreement Likert scale from 1 to 7. Scores were calculated in a Microsoft Excel scoring template provided by the authors for components including active engagement, perceptual abilities, musical training, emotions, singing abilities, and general sophistication. We primarily utilized ‘general sophistication’ as our measure of musicality that took all of the other factors into account. To compare our measure of musical general sophistication to the measure of ‘musical training’ commonly used in previous studies (Białuńska & Dalla Bella, 2017; Chen et al., 2002), we used ‘musical training’ in a second model. Rather than splitting the participants into two groups based upon scoring either high (e.g., > 100) or

low (e.g., < 40) on the Gold-MSI, we used the index as a continuous variable.

Data analysis

To determine how tapping occurred relative to the perfect tempo of the metrical stimulus, we analyzed synchronization accuracy using circular statistics (Fisher, 1995). Each raw data trial was 50 beats long. Each data point was a measure of tapping accuracy, between 0 and ± 300 ms of the tap timing relative to the corresponding beat. Taps with timing outside of these criteria were recoded or removed from the sequence. For example, if any tap occurred greater than 300 ms after the previous cue, it was treated as a tap corresponding to the next cue in the sequence. If a tap occurred less than 300 ms from the previous tap, it was detected as a double tap and removed. If no tap occurred within 300 ms before or 300 ms after a cue, it was detected as a missed tap and removed. These measures were divided by the ISI (600 ms) to convert into relative phase (RP). RP, ranging from -0.5 to $+0.5$ can be represented as vectors around a unit circle (Fig. 1b), as previously utilized (Iversen et al., 2015). Negative (positive) values represent tapping before (after) the target beat, and zero represents perfect accuracy. Accuracy for each trial was quantified as the mean of the 50 separate RP measures. A trial was rejected if we failed to reject the null hypothesis ($p > 0.001$) of circular uniformity on Rayleigh's test, a criterion used in previous studies to suggest random tapping (Iversen et al., 2015; Ruxton, 2017). The mean rejection rate was 3.2% for visual, 1.8% for tactile, and 2.9% for auditory trials.

After rejecting failed trials, we used three measures to quantify tapping performance. The first was the circular standard deviation (SD_{circ}) of tapping asynchronies, defined as $\sqrt{-2 \ln R}$, where R is the mean of the RP vectors. We refer to SD_{circ} as our dependent variable of tapping 'consistency'. A lower SD_{circ} signifies better consistency. We used tapping asynchrony as the second measure, reflecting the average accuracy a trial of taps was from perfect synchrony (zero). A smaller asynchrony (i.e., closer to zero) suggests better accuracy. We ran a linear mixed model with modality (auditory; tactile; visual), deviation level ('full-time'; 'half-time'; 'half + anacrusis'; 'half + acephalic'), and the continuous variable of musical general sophistication (Gold-MSI score) as fixed factors, participant as a random factor, with dependent variable being tapping consistency or accuracy in separate models. We ran model comparisons using Akaike Information Criterion (AIC) scores to determine which combination of fixed factors provided the best fit for our data. A second model contained musical training (Gold-MSI score) instead of musical general sophistication. We predicted worse SMS performance (in both consistency and accuracy) for the visual than for the auditory and tactile

modalities. We also predicted worse performance across modalities as rhythmic deviation increased over deviation levels. Furthermore, better SMS consistency and accuracy were expected from participants with higher musical general sophistication scores.

Autocorrelation of successive taps within the 50-tap time series was our third measurement and was used to reveal possible differences in timing mechanisms underlying SMS performance across deviation levels. Relevant here, a significantly negative autocorrelation for a specific lag (e.g., AC-1) at one rhythmic deviation level, versus a lack of significance at AC-1 for another deviation level, would suggest the presence versus absence of online error correction, respectively (Wing, 2002). The presence or absence of a significant autocorrelation can be determined from the autocorrelation structure, where overall patterns can reflect potential timing strategies participants employ. Thus, we did not take a modeling approach to autocorrelation analysis and instead investigated between-deviation level differences at specific lags for each modality separately. In other words, mean autocorrelation values for each lag were calculated for each trial from its 49 inter-tap intervals (ITIs). 95% confidence intervals (± 0.28) were obtained via the equation $\pm 1.96/\sqrt{N}$, where N was the number of ITIs (49). We analyzed the first four lags, doubling the number of lags we predicted to be significant (Box et al., 1994). Based on previous findings, we predicted significantly negative autocorrelation in auditory and tactile modalities and significantly positive autocorrelation in the visual modality for AC-1 (Hove et al., 2010). Furthermore, we predicted significant AC-3 in the three 'half' deviation levels, suggesting consistent counting in groups of two beats. We also predicted that autocorrelation at higher-order lag AC-4 would not be significant, suggesting the absence of a complex meter with beats in groups of three or more (Iversen et al., 2015). We plotted overall patterns of autocorrelation in each modality and conducted post hoc paired samples *t* tests to identify differences in the autocorrelation function between deviation levels.

All statistical analyses were performed in R statistical software (Team, 2013).

Results

SMS consistency was influenced by modality, rhythmic deviation, and musicality

Linear mixed-effects model comparisons with SMS consistency (SD_{circ}) as the dependent variable revealed a model of best fit (lowest AIC) that included predictor variables modality, deviation level, and musical general sophistication, with the interaction between modality and deviation level, and between deviation level and musical general sophistication.

There were main effects of modality ($F(2, 316) = 37.88$, $p < 0.001$, $\eta_p^2 = 0.19$), deviation level ($F(3, 316) = 37.69$, $p < 0.001$, $\eta_p^2 = 0.26$), and of musical general sophistication ($F(1, 28) = 7.17$, $p < 0.05$, $\eta_p^2 = 0.20$) on SD_{circ} . When the Gold-MSI factor of musical training was used in the model instead of musical general sophistication, the main effect of musical training was not significant ($p = 0.595$).

At the level of modality, Bonferroni corrected post-hoc paired comparisons revealed that SD_{circ} of SMS was significantly smaller (i.e., better consistency) in auditory and tactile modalities than in visual (see violin plot in Fig. 2a). SD_{circ} of auditory (mean = 0.077, SD = 1.017) and SD_{circ} of tactile (mean = 0.080, SD = 0.930) were not significantly different ($t(236) = 0.527$, $p = 0.60$). SD_{circ} for visual (mean = 0.107, SD = 0.945) was significantly higher than for auditory ($t(316) = 3.716$, $p < 0.0001$) and for tactile ($t(237) = 4.487$, $p < 0.0001$), suggesting an overall worse consistency when synchronizing to visual stimuli.

At the level of rhythmic deviation, Bonferroni corrected post hoc paired comparisons revealed that SD_{circ} increased significantly from the ‘full-time’ deviation level to the ‘half-time’ deviation level ($t(316) = -3.421$, $p < 0.005$) and to the ‘half + anacrusis’ deviation level ($t(316) = -4.742$, $p < 0.0001$), suggesting that consistency worsened with rhythmic deviation. However, we found no significant increase in SD_{circ} from the ‘half + anacrusis’ to the ‘half + acephalic’ deviation level ($t(316) = 1.228$, $p = 1.0$).

There was a significant interaction between modality and deviation level ($F(6, 316) = 3.75$, $p < 0.001$, $\eta_p^2 = 0.07$). As seen in Fig. 3, SD_{circ} increased as a function of rhythmic

deviation in all three modalities until the ‘half + anacrusis’ level. While SD_{circ} started to level off from ‘half + anacrusis’ to ‘half + acephalic’ level for auditory and visual modalities, it decreased for the tactile modality. Tukey honest significant difference (HSD) corrected paired comparisons revealed that from the ‘half-time’ to ‘half + anacrusis’ deviation level, SD_{circ} increased significantly for auditory ($t(316) = -3.195$, $p < 0.01$) tactile ($t(316) = -5.368$, $p < 0.0001$), but not significantly for visual ($t(316) = -1.778$, $p = 0.286$). From ‘half + anacrusis’ to ‘half + acephalic’ level, SD_{circ} did not change for auditory ($t(316) = -3.115$, $p = 0.10$) and visual ($t(316) = -2.211$, $p = 0.123$), but significantly decreased for tactile modality ($t(316) = -5.368$, $p < 0.0001$). Overall, the significant interaction was driven primarily by how consistency changed with deviation level in the tactile modality.

Scores for each participant on all Gold-MSI components (active engagement, perceptual abilities, musical training, emotions, signing abilities, and general sophistication) are shown in Supplementary Table 1. Overall, the main effect of musical general sophistication ($F(1, 28) = 7.17$, $p < 0.05$, $\eta_p^2 = 0.20$) on SD_{circ} suggested that a higher musical general sophistication score was associated with a lower SD_{circ} value and thus, better SMS consistency. There was a significant interaction between musical general sophistication and deviation level ($F(3, 316) = 3.78$, $p < 0.05$, $\eta_p^2 = 0.03$). Bonferroni corrected post-hoc tests for association between musical general sophistication scores and SD_{circ} at each deviation level revealed no significant correlation in the ‘full-time’ ($r(28) = -0.28$, $p = 0.129$), ‘half-time’ ($r(28) = -0.37$, $p = 0.042$), or ‘half + anacrusis’ condition ($r(28) = -0.45$,

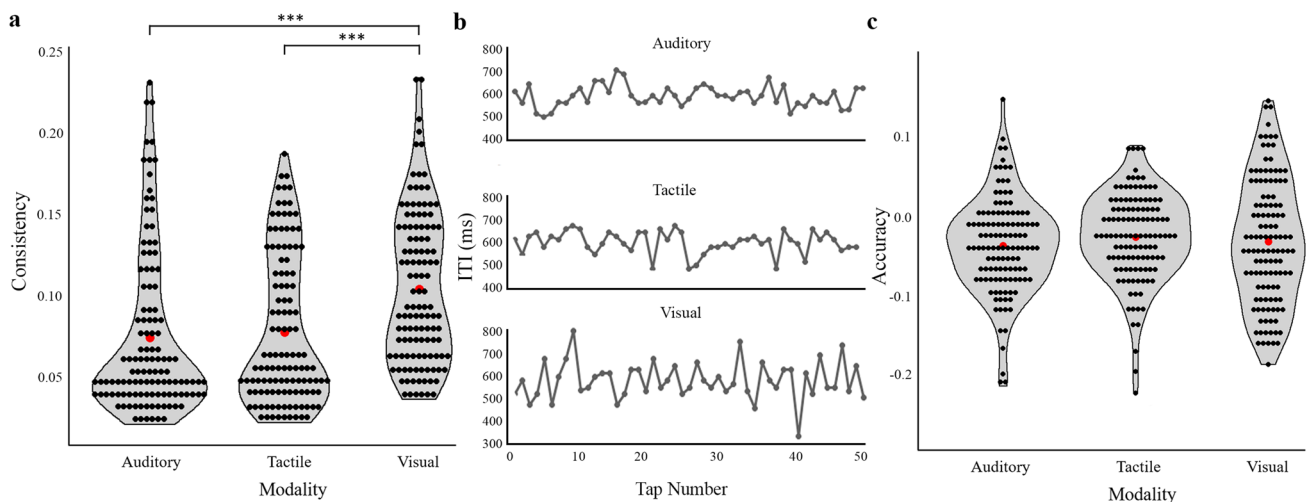
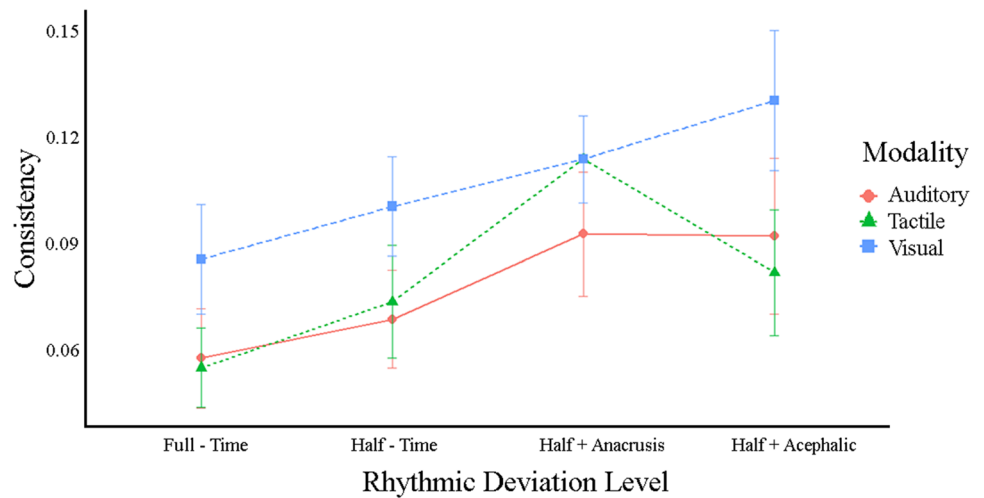


Fig. 2 **a** Violin plot of sensorimotor synchronization consistency in each modality. Consistency (SD_{circ}) is plotted with red dots depicting the mean SD_{circ} . The lower the y-axis, the lower SD_{circ} , and the better the consistency. **b** Examples of tapping time series for each modality. Each ITI is plotted along the x-axis. On the y-axis, 600 ms is the target tempo. Lower values represent tapping too soon, and higher values

represent tapping too late. **c** Violin plot for sensorimotor synchronization accuracy in each modality, with red dots depicting the mean RP. Accuracy is plotted on the y-axis, with negative values representing tapping before and positive values representing tapping after the tempo. Better performance is indicated by accuracy values closer to zero (color figure online)

Fig. 3 Effects of rhythmic deviation level on sensorimotor synchronization for each modality. Consistency (SD_{circ}) is plotted on the y-axis. A lower SD_{circ} represents better SMS consistency. Error bars represent 95% confidence intervals



$p=0.013$). There was a significant negative correlation at the ‘half + acephalic’ deviation level ($r(28) = -0.501$, $p < 0.005$), suggesting that participants with a higher musical general sophistication score had a lower SD_{circ} value (i.e., better SMS consistency).

SMS accuracy was only influenced by rhythmic deviation

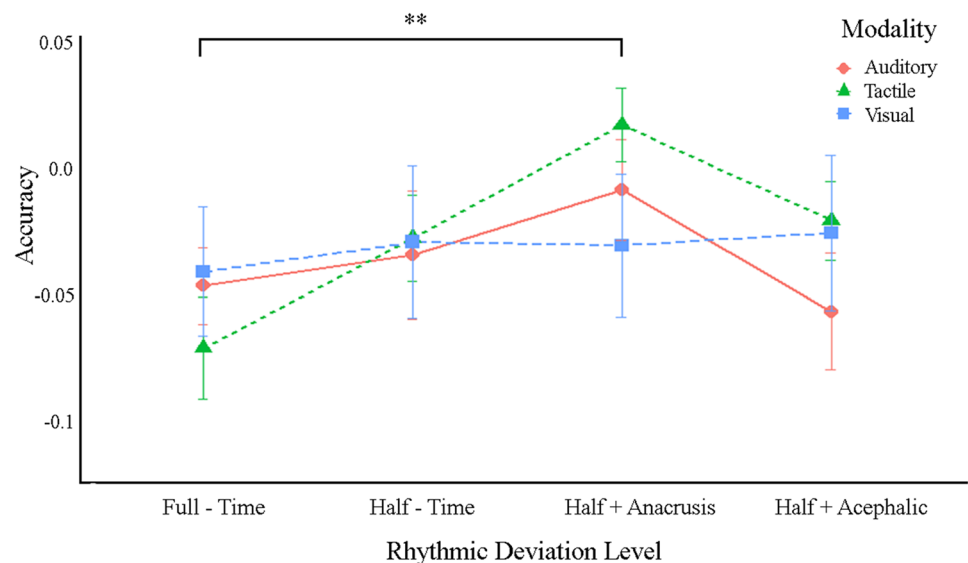
Linear mixed-effects model comparisons with SMS accuracy (mean RP) as the dependent variable revealed a model of best fit (lowest AIC) that included predictor variables modality, deviation level, and musical general sophistication, with the interaction between modality and deviation level, and between deviation level and musical general sophistication. There was no main effect of modality on SMS accuracy ($F(2, 316) = 1.322$, $p = 0.268$). Mean RP for auditory (-0.037 ,

$SD = 0.061$), tactile (-0.026 , $SD = 0.056$), and visual (-0.032 , $SD = 0.079$) were all slightly negative. Hence, participants tapped ahead of the beat across all modalities. There was no main effect of musical general sophistication on SMS accuracy ($F(1, 28) = 0.285$, $p = 0.60$).

There was a main effect of deviation level on SMS accuracy ($F(3, 316) = 11.6$, $p < 0.001$, $\eta_p^2 = 0.10$). As seen in Fig. 4, Bonferroni corrected post-hoc paired comparisons revealed that mean RP negativity in the ‘full-time’ (-0.053 , $SD = 0.041$) was significantly more negative than ‘half + anacrusis’ deviation level (-0.008 , $SD = 0.049$), suggesting less precise accuracy in the ‘full-time’ than in the ‘half + anacrusis’ deviation level ($t(56) = -4.007$, $p < 0.001$). All other paired comparisons did not reach significance (all $ps > 0.03$).

There was a significant interaction between modality and deviation level ($F(6, 316) = 4.052$, $p < 0.001$, $\eta_p^2 = 0.07$). As seen in Fig. 4, accuracy improved (became closer to zero)

Fig. 4 Effects of rhythmic deviation level on sensorimotor synchronization accuracy for each modality. Accuracy is plotted on the y-axis, with negative values representing tapping before and positive values representing tapping after the tempo. Better performance is indicated by accuracy values closer to zero



as a function of rhythmic deviation in both tactile and auditory modalities until the ‘half + anacrusis’ level. Accuracy then worsened from ‘half + anacrusis’ to ‘half + acephalic’ level. Accuracy in the visual modality remained fairly consistent across rhythmic deviation levels. Tukey honest significant difference (HSD) corrected paired comparisons revealed that from the ‘full-time’ to ‘half + anacrusis’ deviation level, accuracy became significantly more precise for tactile ($t(316) = -6.562, p < 0.0001$), but not for auditory ($t(316) = -2.828, p = 0.173$), or visual ($t(316) = -0.765, p = 0.1$) modalities. Overall, the significant interaction was driven primarily by how accuracy changed with deviation level in the tactile modality.

There was a significant interaction between musical general sophistication and deviation level ($F(3, 316) = 3.312, p < 0.02, \eta_p^2 = 0.03$). However, Bonferroni corrected post hoc tests for association between musical general sophistication scores and accuracy at each deviation level revealed no significant correlation in the ‘full-time’ ($r(28) = 0.45, p = 0.013$), ‘half-time’ ($r(28) = -0.06, p = 0.77$), ‘half + anacrusis’ ($r(28) = -0.00, p = 0.98$), or ‘half + acephalic’ ($r(28) = -0.45, p = 0.013$) deviation level.

We conducted a correlation analysis to determine the relationship between SMS accuracy and consistency across modalities and deviation levels. We found a positive correlation ($r(28) = 0.41, p = 0.023$) such that overall better consistency (smaller SD_{circ}) was associated with more precise accuracy (closer to zero mean RP).

Significant autocorrelations across modalities for lag-1 to lag-4

As seen in Fig. 5, we found common patterns in the autocorrelation structure across modalities. Positive

autocorrelations were significant above the 95% confidence interval threshold of 0.283, and negative autocorrelations were significant if below -0.283 . In each modality, there were instances of significant autocorrelations at all four lags.

Auditory AC-1 means were significantly negative for ‘half-time’ ($-0.293, SD = 0.125$), ‘half + anacrusis’ ($-0.395, SD = 0.183$), and ‘half + acephalic’ ($-0.507, SD = 0.15$) deviation levels (Fig. 5a). The means of auditory AC values at higher-order lags were significant only for the ‘half + acephalic’ deviation level, with AC-2 being significantly positive ($0.323, SD = 0.19$), AC-3 being significantly negative ($-0.396, SD = 0.146$), and AC-4 being significantly positive ($0.347, SD = 0.158$). Tactile AC-1 means were significantly negative for ‘half + anacrusis’ ($-0.341, SD = 0.192$) and ‘half + acephalic’ ($-0.42, SD = 0.192$) deviation levels (Fig. 5b). In addition, AC-3 mean was significantly negative for the ‘half + acephalic’ deviation level ($-0.324, SD = 0.186$). Both visual AC-1 and AC-3 means were found significantly negative, only for the ‘half + acephalic’ deviation level ($-0.426, SD = 0.134$ and $-0.288, SD = 0.185$, respectively, Fig. 5c).

Using a corrected p -value of 0.008 ($0.05/6$), we conducted t tests to compare significant AC-1 means between deviation levels for each modality separately. We focused only on comparisons involving deviation levels with significant AC-1 values. No significant difference was found between ‘half + anacrusis’ and ‘half + acephalic’ for auditory ($t(56) = 2.582, p = 0.012$) or tactile ($t(56) = 1.737, p = 0.088$). Similarly, we did not find a significant difference between auditory ‘half-time’ and ‘half + anacrusis’ AC-1 means ($t(56) = 2.542, p = 0.014$). However, the auditory ‘half + acephalic’ deviation level AC-1 mean was significantly more negative than the auditory ‘half-time’ deviation level ($t(56) = 6.014, p < 0.0001$).

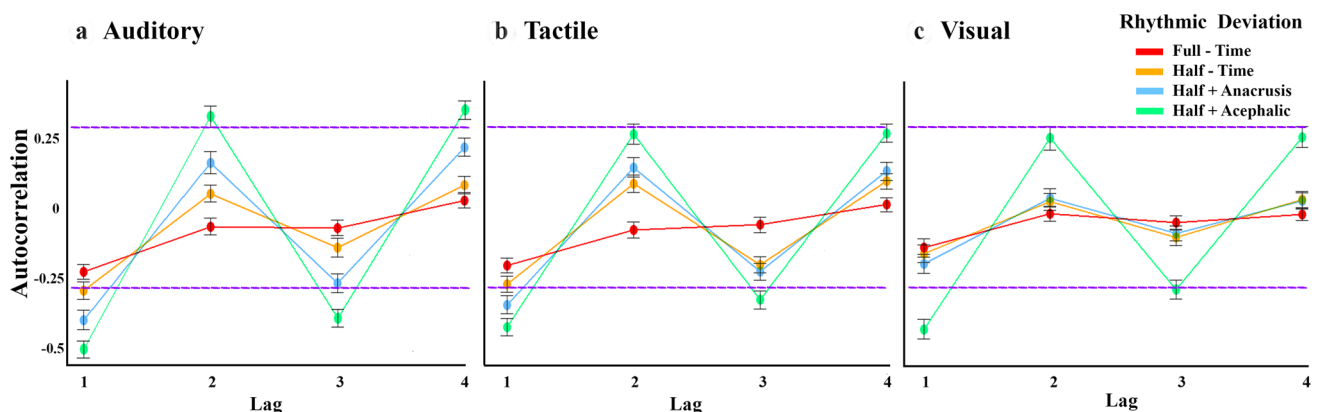


Fig. 5 Autocorrelation of tapping time series for the four rhythmic deviation levels in **a** Auditory, **b** Tactile, and **c** Visual modalities. Autocorrelations above the top dashed line (representing the upper

bound of 95% confidence interval: 0.28) or below the bottom dashed line (representing the lower bound of 95% confidence interval -0.28) are significant. Error bars represent the standard error of the mean

Discussion

The main aim of the current study was to establish factors that impact SMS in the general population. We used a finger-tapping task and manipulated rhythmic deviation of the stimulus identically in auditory, tactile, and visual modalities. From the tapping time series in each trial, we utilized relative phase (RP) as a measure of SMS accuracy of temporal stimulus perception. The circular standard deviation (SD_{circ}) of tap timing for the trial provided a broader measure of SMS consistency. We also quantified the autocorrelation of the inter-tap-interval (ITI) time series to reveal potential time-keeping strategies. We administered the Gold-MSI questionnaire to determine our participants' musical familiarity and aptitude.

SMS consistency is influenced by modality, rhythmic deviation, and musicality

At the level of modality, we found that SMS consistency to external rhythmic stimuli was significantly better for auditory and tactile than for visual flashing sequences, as previously found (Patel et al., 2005; Repp & Penel, 2004; Varlet et al., 2012). When asked after the procedure which modality they found most difficult, participants unanimously chose the visual modality as the most challenging, regardless of the order of modalities they were randomly assigned to. However, in contrast to previous studies reporting an inability to synchronize with flashes (Patel et al., 2005), our participants were able to synchronize with visual stimuli.

We found that SMS to visual stimuli was influenced by task difficulty (increased rhythmic deviation) in a similar way to auditory, and to an extent, tactile. Notably, the source of interaction between modality and rhythmic deviation was driven by how consistency (SD_{circ}) was changed by the 'half + anacrusis' deviation level in the tactile modality. Consistency improved from the 'half + anacrusis' to the 'half + acephalic' level for tactile modality, whereas minimal change occurred in auditory and visual modalities. Participants unanimously commented that they found the intensity difference of the vibrations more difficult to interpret than the beeps. It is not clear why participants synchronized better to the pattern with tactile intensity differences when presented with the acephalic versus being presented with the anacrusis. Nevertheless, how consistency changed from the 'half + anacrusis' to the 'half + acephalic' level in the tactile modality might contribute to the non-significant difference between these two levels across modalities. Despite the importance of psychophysical differences in stimulus intensities (Merchel & Altinsoy, 2020), how they affect SMS performance is beyond the scope of the current study.

Contrary to previous studies that support SMS to auditory stimuli as being inherently different from that of SMS to visual stimuli (Patel et al., 2005; Repp & Penel, 2004), we found that the influence of rhythmic deviation was similarly experienced in both modalities. This similar effect of rhythmic deviation across modalities was apparent despite our discussed finding of significantly worse consistency of SMS performance to visual than to auditory and tactile stimuli. The inclusion of an introductory phase in each trial might have played a role in this finding. The introductory phase was given to allow the participant chance to familiarize themselves with the rhythm before recording their results, of which they were made explicitly aware. Previous studies have eliminated the first few taps before analyzing the results (Iversen et al., 2015) due to the common finding that synchronization requires a few taps to become stable (see Junkai et al., 2020 for a meta-analysis). The introductory phase intuitively increases familiarity and comfortability in synchronizing with the given rhythm. Hence, including the introduction serves two functions: (i) an anticipatory function that might encourage engagement of sensory predictions by the motor system (Iversen & Balasubramaniam, 2016); and (ii) a reactive function that encourages dynamic timing strategies (Van Der Steen & Keller, 2013). Future studies could further examine the effect of including an introductory synchronization period.

Alternatively, SMS performance to visual stimuli cannot be fully interpreted without accounting for possible contributions of auditory imagery. As our most temporally salient modality, and one that most individuals can instantiate in their minds at will (Hinwar & Lambert, 2021), auditory imagery abilities might assist with SMS in other non-auditory modalities. Using fMRI, it has been suggested that auditory imagery is automatically activated and persistently influential while a visual rhythmic sequence is being observed (Grahn et al., 2011). If the automatic transformation of visual information into sound is taking place while participants synchronize with visual cues, it might have contributed to the preserved effect of rhythmic deviation despite a significant overall cost to consistency.

We found a significant negative correlation between musical general sophistication scores and SD_{circ} in our most rhythmically deviant level, suggesting that the higher one's musical general sophistication, the better one's SMS consistency. This held for all three modalities. However, this was not true when we replaced our musical general sophistication measure with the measure of musical training, i.e., years of formal training on a musical instrument. Musical general sophistication might be a more sensitive measure in future studies examining the influence of musicality on SMS performance, due to the finding that scores positively correlate with personality traits of extroversion and openness (Müllensiefen et al., 2014). Traits of high extraversion

and openness may suggest one's tendency to synchronize with external stimuli in the social environment. SMS skills have been found to generalize to skills of speech pattern detection (Lagrois et al., 2019). Being able to predict the timing of events is essential for relating to other people and articulating oneself in the social environment (von Schnehen et al., 2022). Therefore, high extraversion and openness in individuals with higher general sophistication scores may contribute to their better SMS consistency in the present study. Note that there was no significant correlation between musical general sophistication scores and SD_{circ} in the other three levels of rhythmic deviation, possibly due to a ceiling effect in these levels.

SMS accuracy was only influenced by rhythmic deviation

We observed that accuracy (mean RP) did not differ between modalities, suggesting that perception of where beats occurred was similarly experienced. This is a rare finding since many studies report significantly less accurate SMS performance for discrete visual stimuli than for auditory (Lorås et al., 2012; Patel et al., 2005; Repp & Penel, 2002; Varlet et al., 2012); especially when the inter-stimulus-interval (ISI) becomes less than 500 ms (Repp, 2003), or the visual sequence included subdivisions of flashes (Patel et al., 2005). However, Patel et al. (2005) suggested that beat induction was just as effectively accomplished in the visual modality when a sufficient effort is made to focus on the underlying beat. In addition to the potential contribution of large individual differences in SMS accuracy (Aschersleben, 2003), there might be a contribution from procedural differences: We used visual and auditory stimuli of the same length (33.3 ms), whereas previous studies reported significantly reduced SMS accuracy with visual stimuli but used visual and auditory stimuli of different lengths (e.g., 100 ms flash and 50 ms beep in Iversen et al., 2015). Less accurate visual SMS performance was also reported in Chen et al. (2002) when an ISI of 488 ms was used. Differences in stimulus length or the use of a suboptimal ISI duration might influence SMS accuracy between modalities.

We did find a significant effect of deviation level on accuracy across modalities: compared to 'full-time', the negativity of the mean RP values became smaller (i.e., more accurate) in the 'half + anacrusis' deviation level, suggesting higher accuracy (and lesser tendency to actively predict the stimulus). This finding is somewhat surprising given the 'half + anacrusis' level contains more deviation than 'full-time'. The more accurate performance in the 'half-anacrusis' level may be due to the presence of 'pick-up' cues. Although its presence can be distracting, 'pick-up' cues can bridge the gap to the main cue and potentially improve accuracy. The source of interaction

between modality and rhythmic deviation was primarily driven by how accuracy improved between the 'full-time' and 'half + anacrusis' deviation levels in the tactile modality. Perhaps the 'pick-up' cue is more effective in the tactile modality than in the auditory and visual modalities. Notably, while SMS accuracy in the tactile modality was most precise for the 'half + anacrusis' deviation level, this was also the deviation level performed with the least consistency (see Section "SMS consistency is influenced by modality, rhythmic deviation, and musicality"). Our results might suggest a possible trade-off between accuracy and consistency for the tactile modality, where the reliance on the 'pick-up' cue in the tactile stimulus pattern improved accuracy at the cost of consistency.

Our findings of an overall negative mean RP and a positive correlation between overall accuracy and consistency of SMS performance are consistent with previous work (See Junkai et al., 2020 for a meta-analysis). Accuracy and consistency of SMS performance represent different components of the process, detection of regularities in the stimulus, and the ability to keep synchronizing for a considerable length of time, respectively (Nozaradan et al., 2016). It is proposed that the correlation between accuracy and consistency is not commonly investigated in studies due to a lack of a theory or evidence of a causal relationship between them (Junkai et al., 2020).

Our lack of a musical general sophistication effect on SMS accuracy does not replicate previous findings that negative asynchrony (negativity of mean RP values) was significantly reduced in those with musical training (Aschersleben, 2002). Furthermore, our analysis did not reveal the source of interaction between musical general sophistication scores and rhythmic deviation (i.e., no significant correlations between scores and SMS accuracy at any rhythmic deviation level used in the current study). Our findings are consistent with reports of large individual differences in SMS accuracy (Repp & Penel, 2002), and that only participants with specifically rhythm-based training were expected to display a significant reduction in negative asynchrony (Manning & Schutz, 2016). For example, dancing programs have been shown to improve SMS accuracy in young children (Kapidistria et al., 2021; Chatzihidirolou et al., 2018). However, whether better SMS accuracy can be achieved as a direct result of musical training therefore might be influenced by other factors. Slater et al. (2013) found that accuracy improvement was only observed during the continuation phase of trials, where finger-tapping was performed without the guiding external stimulus. Hence, SMS accuracy might improve via the development of auditory working memory and imagery skills (Strait et al., 2012). Furthermore, it has been reported that a positive relationship between auditory imagery skills and SMS accuracy was only partly mediated by music training (Pecenka & Keller, 2009).

Significantly autocorrelations across modalities for lag-1 to lag-4

Our finding of significant negative autocorrelations at AC-1 was expected for auditory and tactile modalities. Negative AC-1 is a common finding for SMS tasks where an auditory or tactile stimulus follows a simple isochronous beat (Ammirante et al., 2016; Guérin et al., 2021). The use of a predictive autocorrection strategy, as reflected by a negative AC-1, was most prominent in the ‘half + acephalic’ level. The lack of significant autocorrelation values at higher-order lags for ‘full-time’, ‘half-time’, and ‘half + anacrusis’ deviation levels suggests that participants only counted metronomically during these trials. In contrast, during ‘half + acephalic’ deviation level trials, significant autocorrelations at higher-order lags (specifically the negative AC-3) indicate the metrical time-keeping signature of counting in a binary fashion (Iversen et al., 2015; Vorberg & Hambuch, 1978). This more complex strategy might be driven by the tendency to concentrate synchronization on every second beat in the pattern. The second beat did not have a guiding stimulus but might have been preferable to focusing on the combination of the first beat and acephalic. Surprisingly, this was apparent for all modalities. Hence, regardless of modality, the same strategy was used for the most challenging rhythmic deviation level. Our results contradict findings that higher-order metrical strategies are found for auditory tones, but not for visual flashes (Iversen et al., 2015), suggesting that there might be a modality-independent facility to engage metrical time-keeping strategies (Chen et al., 2002).

Note that for visual flashes, a predictive autocorrection strategy (negative AC-1) was not apparent in ‘half-time’ and ‘half + anacrusis’ conditions. However, we did find negative AC-1 for visual flashes at the most rhythmically deviant ‘half + acephalic’ condition, contrary to previous findings that SMS to visual flashes typically elicits a positive AC-1 (Hove et al., 2010). A positive AC-1 suggests emergent timing, a process that is distinctly different from predictive autocorrection (Guérin et al., 2021). Emergent timing is a feature previously reported for SMS to visual flashes but not for auditory tones or tactile vibrations, and is suggested to represent worse synchronization (Hove et al., 2010). It is unclear what drives a participant to use either a predictive or an emergent timing strategy. It has been found that attention and cognitive control can be adapted to facilitate better SMS performance when tempo changes are introduced into an auditory sequence (Repp & Keller, 2004). Our observation of predictive autocorrection in our most rhythmically deviant condition for the visual modality might suggest that timing strategies could be dependent on changes in attentional focus as a result of increased task difficulty. Nevertheless, the lack of predictive autocorrection strategy in two of our less rhythmically deviant conditions could have contributed

to our finding of overall lower SMS consistency for the visual modality.

Summary

The current study examined the effect of stimulus modality, rhythmic deviation, and musical general sophistication on SMS performance. For SMS consistency, we found that the modality of the stimulus had a significant effect, with better consistency for auditory and tactile than for visual modality. A significant influence of rhythmic deviation on SMS consistency was found for all three modalities, with worse consistency as rhythmic deviation increased. An interaction between modality and rhythmic deviation was driven by how the consistency was differently influenced by the ‘half + anacrusis’ deviation level in the tactile modality. Lastly, we observed a significant effect of musical general sophistication on SMS consistency, where better SMS consistency in the most deviant condition was observed in participants with higher musical general sophistication scores. For SMS accuracy, rhythmic deviation of the stimulus was the only factor that had a significant influence, with more precise performance in the ‘half + anacrusis’ condition than in the easier ‘full-time’ condition.

Limitations and future directions

As discussed above, while eliminating the first few taps of the trial is a more common methodological approach (Iversen et al., 2015), we included an introduction phase before each trial to minimize data rejection of initial performance errors. Task features, such as an introduction phase might have made the participants more comfortable and therefore minimized the contribution of a predictive timing strategy, as has been previously suggested (Delignières & Torre, 2011). However, we did not test this hypothesis. Future studies could examine how timing strategies are influenced by the inclusion of an introduction preceding synchronization to external stimuli. Furthermore, we have speculated that the use of a specific timing strategy might be dependent on task difficulty induced by different levels of rhythmic deviation. Alternatively, increased task difficulty might be a subjective feature inversely related to one’s level of musicality. The effect of musicality on SMS timing strategies can be further elucidated using higher levels of rhythmic deviation than those in the current study. Lastly, SMS performance, auditory imagery ability, and musicality have been found to all positively correlate in a sample of musicians, suggesting that there is a relationship between musicality and auditory imagery abilities (Pecenka & Keller, 2009). Future studies can further examine the role of

auditory imagery and how it interacts with musicality using paradigms that actively inhibit the use of auditory imagery in SMS.

Conclusions

We found that SMS to external rhythmic stimuli was more consistent for auditory and tactile than for visual sequences. Further, we found that SMS performance significantly decreased in all modalities as rhythmic deviation increased. The higher one's musical general sophistication score, the greater one's SMS performance in the most difficult deviation level. Combined, these findings suggest that rhythmic synchronization performance is affected not only by the modality and rhythmic deviation of the stimuli but also by the musical general sophistication of the synchronizer.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00426-023-01801-3>.

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Data availability The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent Informed consent was obtained from all individual participants included in the study.

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